



Effect of fibre orientation on the light scattering during laser transmission welding

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ARTICLE INFO

Keywords:

Laser transmission welding
Composite material
Ray tracing method
Light scattering

ABSTRACT

During laser welding of thermoplastic (LTW) composites reinforced with short fibres, a divergence of the laser beam is observed due to internal refraction of the beam at each matrix–fibre interface. This phenomenon leads to scattering of the laser beam in this heterogeneous medium, resulting in reduction of energy reaching the weld interface. This work presents a numerical study of the effect of fibre orientation in fibre-reinforced composites during the laser transmission welding simulation. A three dimensional (3D) numerical structure is generated to take into account the real microstructure of a composite material reinforced with short fibres. The information of fibre volume fraction, fibre length distribution and fibre orientation distribution is extracted from injection moulding simulations. Fibre orientation distribution function is extracted from fibre orientation tensors. An algorithm is implemented to trace rays propagating through composite material at two scales: microscopic and macroscopic. This algorithm uses optical properties of the composite material to simulate laser beam reflection and refraction in a complex structure. Laser beam scatter at the weld interface is simulated in 3D geometries with various fibre orientation distributions. The effect of fibre orientation on the light scattering phenomenon of laser is compared for different cases of fibre orientation. The simulation results are validated experimentally.

1. Introduction

Laser transmission welding is a popular joining process for thermoplastic parts in automotive, medical and electronic sectors [1], especially when the control of heat source location, size and magnitude is critical [2]. If this process is to be applied to two compatible [3] thermoplastic parts, one of these parts should be transparent to laser wavelength and the other absorbent to the same wavelength. The two parts are clamped together and a laser beam is projected towards their interface. The laser beam is transmitted through the transparent part and is absorbed at the surface of the absorbent part, where the absorbent part melts locally. At this weld interface, the transparent part also melts locally due to heat conduction (see Fig. 1). A bond is formed upon solidification. When fibres are included in the transparent part, it becomes semi-transparent and diffusive.

The quality of the weld depends on the interface weld temperature which is impacted by the optical and thermal properties of the composite parts and the welding parameters [4]. During its propagation in a heterogeneous medium, such as fibre reinforced composites, the laser beam undergoes reflection and/or refraction at every fibre–matrix

interface in their path. This leads to the scatter of laser rays before reaching the weld interface causing a reduction in laser intensity, and hence attenuation in energy reaching the interface [5,6]. It has been reported that the temperature distribution at the weld seam strongly depends on the local fibre orientation [7]. To predict the welding performance of a fibre reinforced thermoplastic composite, it is necessary to analyse the influence of fibre orientation on the laser transmission.

The mechanical properties of composite materials are a function of fibre orientation. Various works [8,9] have been done to better predict the influence of orientation distribution of fibres on the mechanical behaviour of materials. Manufacturing process conditions and matrix flow patterns can be linked to fibre orientation distribution in short fibre reinforced thermoplastic (SFRT) composites. In this work, the reflection and refraction of light transmission through SFRT composites are studied to predict the influence of fibre orientation on laser transmission welding of SFRT.

During the injection moulding, flow conditions determine the orientation of short fibres [10]. A common feature of fibre orientation seen in injection moulded parts is the shell–core distribution. The orientation of

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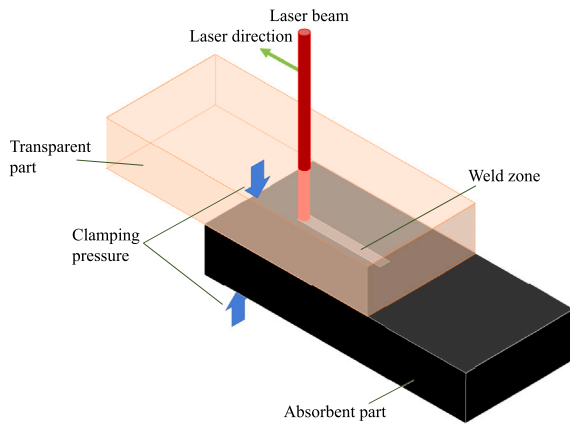


Fig. 1. Schematic of laser transmission welding.

fibres at the part shell is parallel to the flow, whereas it is perpendicular at the core. Highly aligned fibre orientation is also observed at the inlet and exit of a mould. The direction of fibres within the material strongly impacts the mechanical properties [8,11–13]. In this work, it is shown that it also influences the weld seam properties.

Qualitative measurements of fibre orientation have been carried out by Micro-Computer Tomography (μ CT) [14], Fraunhofer light scattering [15], confocal techniques [16], image analysis [17] etc. All these approaches have commonly used fibre orientation tensors [18] to describe the fibre orientation distributions. The present work aims to investigate the deviation behaviour of the light transmission according to the fibre orientation.

In this paper, the dependency of light diffusion patterns on the fibre orientation is discussed. Firstly, three dimensional numerical representative volume elements of composite materials are constructed by varying fibre orientations. The fibre orientations are represented using fibre orientation tensors. Ray tracing simulations are performed subsequently, to model the diffusion patterns of the light transmitting through these RVEs. Profiles of the scattered rays are generated at the weld interface and are compared for different fibre orientation states. The simulation results are validated by comparison with experimental results.

2. Construction of 3D numerical geometry

To model the diffusion of a laser beam considering the orientation of short fibres, 3D RVEs of a semi-transparent part should be constructed. 3D numerical geometries based on the observations of a real SFRT composite material are constructed by an algorithm developed on MATLAB. This algorithm takes into account fibre morphology such as fibre orientation distribution, fibre shape, fibre volume fraction and fibre aspect ratio.

The algorithm loop chooses a random coordinate as the centre of a fibre inclusion in a cube cell with a regular 3D grid. In the beginning, all points of this grid can be chosen as centres of inclusions. The first fibre inclusion is placed at this centre after allocating a fibre aspect ratio and orientation properties to it. The grid coordinates occupied by this fibre inclusion are no more considered for placement of future fibre inclusions. The next fibre inclusion is placed by temporarily eliminating all the grid points occupied by this fibre inclusion if it is to use the coordinates occupied by the past fibre inclusions as the centre. A final centre for this inclusion is chosen randomly from the remaining coordinates. The placement of the inclusions in the geometry is done with level sets. This approach (as seen in Fig. 2(a)) avoids overlapping of the inclusions in the 3D grid as in Poisson's process. This cell is also generated with periodicity (Fig. 2(b)).

Three dimensional geometries are constructed according the following criteria.

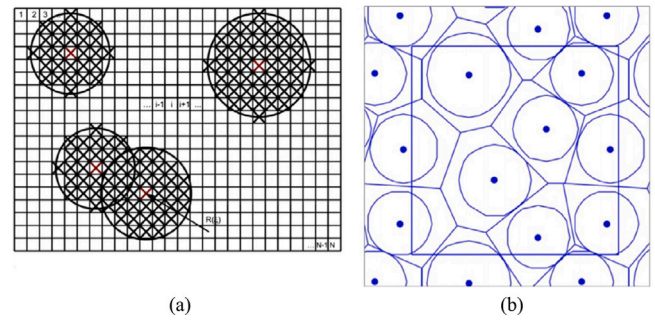


Fig. 2. Grid used in algorithm showing (a) inclusion interaction (b) inclusion centres and periodicity [19].

2.1. Fibre shape

The model creates polyhedrons to represent the fibres in a 3D grid. The algorithm uses the parametrisation of an ellipsoid to define the inclusions. Several geometric shapes can be included in the geometry, such as ellipsoids cylinders, cuboids or regular polyhedrons. As the algorithm allows regular polyhedrons to represent fibre inclusions in the numerical geometry, it permits representation of complex shapes of natural fibres in the geometry with simple modifications. To simplify the simulations in this paper, the fibre shape is selected to be cylinders as are glass fibres [20].

2.2. Fibre length distribution

The information on the fibre length distribution in a reference was used in this work. Based on injection moulding simulations of short fibre composites on Autodesk MOLDFLOW, a fibre length and radius analysis was performed by Soete et al. [21]. The average length of the fibres was 500 μ m and the radius was 100 μ m. This data is used to define a probability distribution function to assign the length of the fibres in the grid.

2.3. Obstruction of the laser beam during transmission

During the numerical simulations, a laser beam is discretised into many rays to trace their paths. Every ray that travels through the numerical geometry should be obstructed at least once in its path to avoid a ray concentration effect. From the view point of the laser source, the generated geometry can be visualised using image processing. In a monochrome image (Fig. 3), two regions can be evidently distinguished: the black area represents the fibres, and the white area represents the region where the laser beam can travel through the numerical geometry without any obstructions in its path. The intention of this criterion is to construct the RVE while minimising the white areas, to obtain the direction of ray path deviation by imposing at least one interaction between fibres and laser. White lines shown in Fig. 3 are the outlines of fibres which are represented for the sake of illustration.

2.4. Fibre volume fraction

The numerical geometry construction converges to about 99% obstruction (black region) and cannot reach 100% without significantly increasing the fibre volume fraction. To reduce the computation cost, the convergence is preferred over achieving a full obstruction to ray paths. This also risks in allowing a few rays to travel uninterrupted to the weld interface and these rays generate a local heat concentration at the weld interface. This circumstance is also expected, however, during light diffusion with complete obstruction. The fibre volume fraction converges at 29% maintaining the convergence of black region.

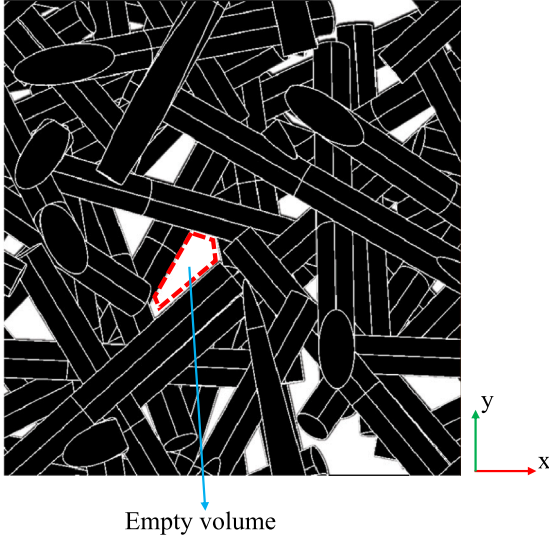


Fig. 3. Imposition of obstruction to every ray path.

2.5. Fibre orientation distribution

The prediction of fibre orientation state using orientation distribution function which is a kind of a probability density function, by F. Folgar and C. Tucker [8] and its extension into use of Fibre Orientation Tensor (FOT) by S. Advani and C. Tucker [18] have been a common approach to investigate the influence of fibre orientation on thermal and mechanical properties of composite materials [22].

The orientation of the fibre inclusions in the numerical geometries is defined using rotation matrices with respect to axes 1 and 3. These angles of rotation are ϕ and θ , and are obtained from fibre orientation tensors. FOTs are simplified descriptions of orientation distribution function defined by the orientation of a unit orientation vector p with respect to coordinate axes as shown in Fig. 4. A second order FOT is written as

$$a = \begin{bmatrix} \langle p_1 p_1 \rangle & \langle p_1 p_2 \rangle & \langle p_1 p_3 \rangle \\ \langle p_2 p_1 \rangle & \langle p_2 p_2 \rangle & \langle p_2 p_3 \rangle \\ \langle p_3 p_1 \rangle & \langle p_3 p_2 \rangle & \langle p_3 p_3 \rangle \end{bmatrix} \quad (1)$$

and hence, the components of orientation of p are written as

$$p_1 = \sin \theta \cos \phi \quad (2a)$$

$$p_2 = \sin \theta \sin \phi \quad (2b)$$

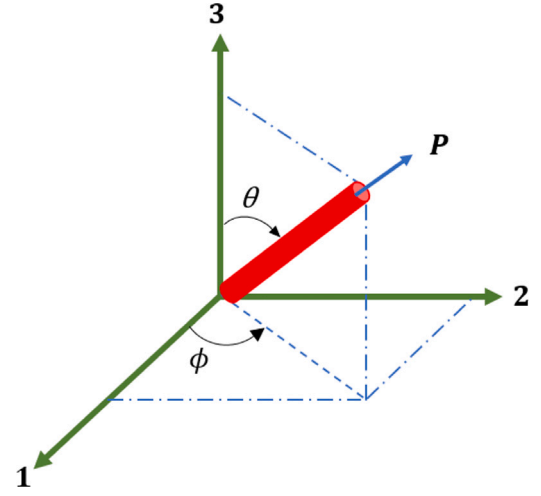
$$p_3 = \cos \theta \quad (2c)$$

The orientation vector p can be used to describe the orientation distribution function, ψ with deviatoric versions (b_{ij} , b_{ijkl}) and tensor basis functions (f_{ij} , f_{ijkl}).

$$\psi(p) = \frac{1}{4\pi} + \frac{15}{8\pi} b_{ij} f_{ij}(p) + \frac{315}{32\pi} b_{ijkl} f_{ijkl}(p) + \dots \quad (3)$$

Fibre orientation tensors hold information of orientation described by the orientation distribution function (ODF). The set of all possible directions of p corresponds to a unit sphere. A second order tensor is a set of nine elements out of which only six are unique due to symmetry [23].

Autodesk MOLDFLOW generates second order orientation tensors in each numerical element of the injection moulded plate. A drawback of the second order tensor is the low effectiveness of translating the tensor into an informative ODF. It should be noted that at higher the order of the tensor, higher is the accuracy of the fibre orientation [24]. Fig. 5

Fig. 4. Definition of orientation of unit vector p with angles θ and ϕ .

shows that increasing the order of the FOT increases the accuracy. Hybrid closure approximation (a_{ijkl}^{HYB}) [25] is widely used to extend the second order orientation tensor to the fourth order in numerical simulations of injection moulding process [13,26]. This closure approximation is a sum of two other closure approximations: linear (a_{ijkl}^{LIN}) and quadratic (a_{ijkl}^{QUA}).

$$a_{ijkl}^{HYB} = (1 - f) a_{ijkl}^{LIN} + f a_{ijkl}^{QUA} \quad (4)$$

where, $f = 1 - 27 \det(a_{ij})$.

It has been observed that when using hybrid closure approximation, around 1% of the orientation angles θ and ϕ is out of the expected range of orientation. These samples can be seen in Fig. 5(b) as a vertical orthodrome. These angles are not eliminated during geometry generation, in this work. Von Neumann rejection algorithm [27] is used to sample θ and ϕ from the fourth order ODF. A composite microstructure is generated from fibre orientation tensors using the algorithm implemented in MATLAB (See Fig. 6).

The algorithm is verified by sampling many pairs of orientation angles and constructing a composite geometry by randomly selecting pairs from the samples. Subsequently, orientation angle information is collected from the fibre inclusions in the geometry and an FOT is regenerated from it (Fig. 6). Comparing the resulting fibre orientation tensor with the initial input fibre orientation tensor, it can be confirmed that more than 99% accuracy is achieved and the method is verified.

3. Ray tracing method

The strength of weld (σ) formed during laser transmission welding is a function of clamping pressure [28], welding temperature and healing time [29,30].

$$\sigma = f(p, T, t) \quad (5)$$

The temperature in the weld area should be above glass transition temperature (for amorphous polymers) or melting temperature (for semi-crystalline polymers) of the involved materials to reduce local viscosity and allow inter-molecular inter-diffusion of polymer chains [30]. The focus of this work is the welding temperature because the change in shape and intensity of the laser beam at the weld interface influences the temperature field that forms a weld [31,32]. Once the laser beam enters the transparent part, the optical properties of the semi-transparent part (ratio of refractive indices of the matrix and the fibres)

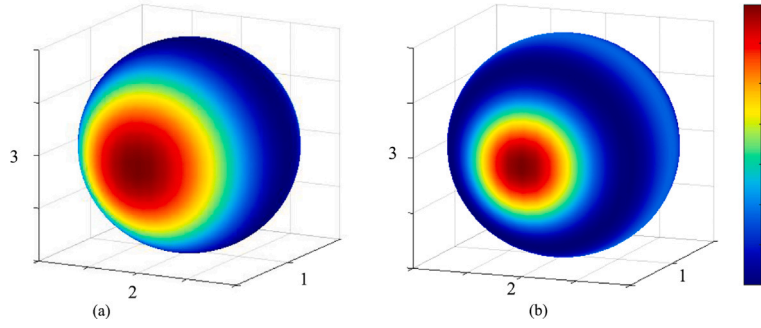


Fig. 5. Orientation distribution of fibre shown in a unit spheres, generated from fibre orientation tensor with components $a_{11} = 1$ and $a_{22} = a_{33} = 0$. (a) second order ODF (b) fourth order ODF.

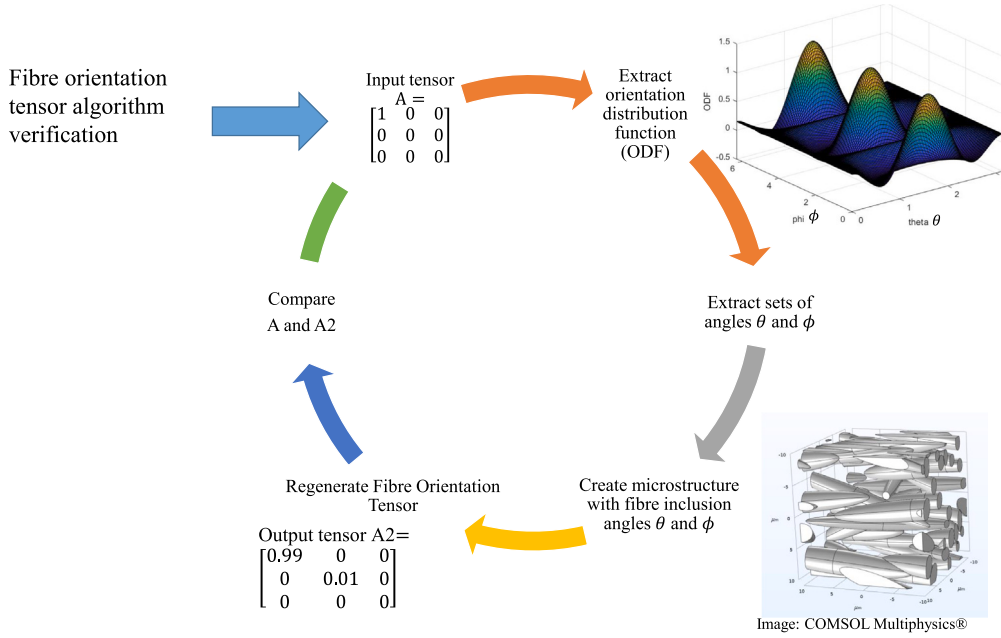


Fig. 6. Method verification to use fibre orientation tensors to define orientation distribution function.

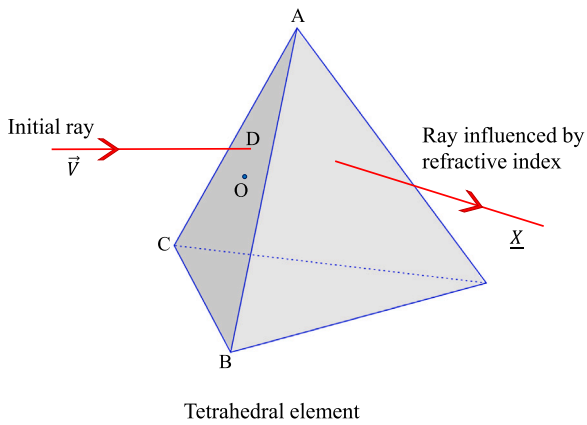


Fig. 7. Ray tracing simulation at micro scale.

play a vital role in directing the ray path [33]. Hence, the direction of propagation of laser rays is of interest.

The ray tracing method is based on geometrical optics. In a complex medium such as a composite structure, the propagation of laser rays can be simulated using ray tracing method which is based on the

Snell–Descartes law [31].

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \quad (6)$$

where, n_i and θ_i ($i = 1, 2$) are the refractive indices of the medium i and the angle formed by the ray and normal at the interface. The circular laser beam is discretised into rays which are distributed in a Gaussian or square form.

The previously constructed 3D geometry is meshed with finite element method. In this heterogeneous medium, the change in path and direction of all laser rays is calculated at each instance of change in the medium at a micro-scale. Each ray interacts with every element in its individual path and changes the direction based on the change in refractive index if the medium is alternated.

When there is a change in the medium, the tetrahedral element at the intersection plays a role to change the ray direction. The ray enters through one face and its exit face is calculated based on the ratio of refractive index of the media involved. To find this exit face, the ray tracing simulation tests all the faces of the tetrahedron element which the ray has entered. For the triangular face (ABC), in an orthonormal frame of reference with any centre O , and a ray entering from point D and direction vector $\vec{V}$, there is intersection when the following condition is reached (See Fig. 7).

$$u \leq 1 \& u \geq 0 \& v \leq 1 \& v \geq 0 \& u + v \leq 1 \& u + v \geq 0 \& t > 0 \quad (7)$$

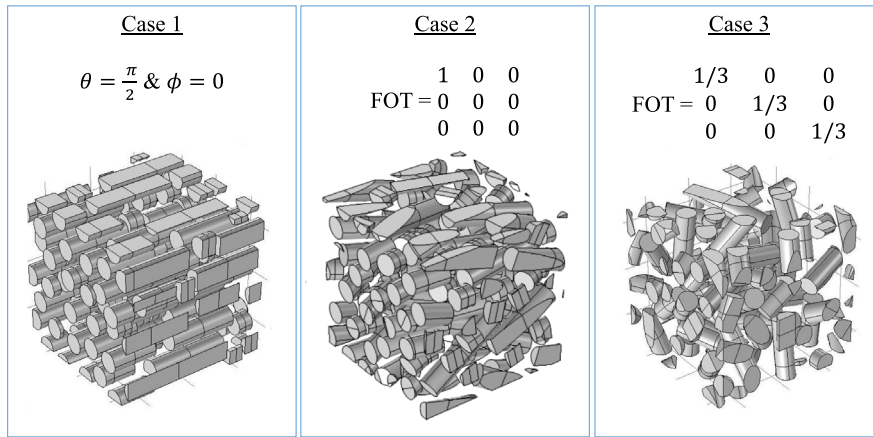


Fig. 8. Three cases of geometries with varying fibre orientation.

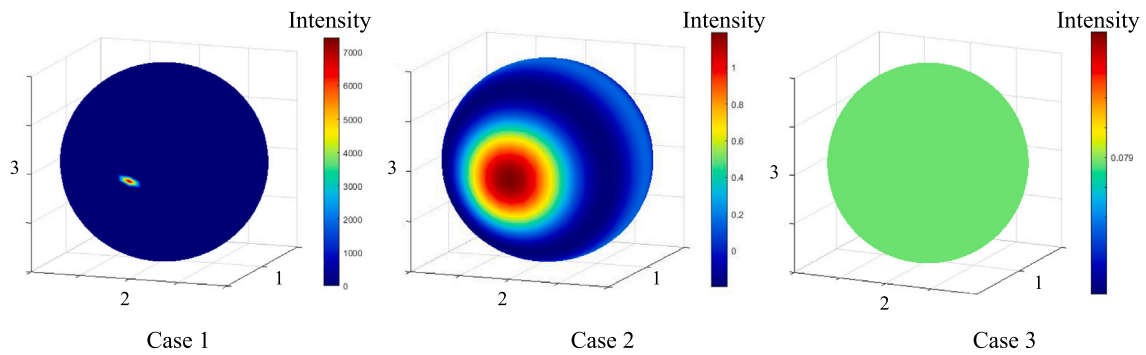


Fig. 9. Orientation distribution of fibres in 3 cases of geometries.

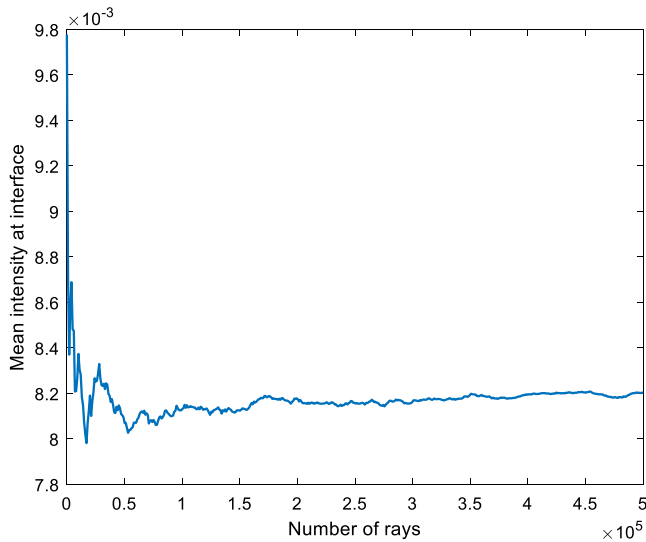


Fig. 10. Convergence of number of rays.

$$\underline{X} = \begin{bmatrix} u \\ v \\ t \end{bmatrix}$$

$$\underline{X} = [\overline{OB} - \overline{OA}, \overline{OC} - \overline{OA}, -\vec{V}] \setminus (\overline{OD} - \overline{OA}) \quad (8)$$

where u, v, t are the components of the vector $\underline{X}$ which are calculated by Eq. (8). The new starting point D is calculated as $\overline{OD} \rightarrow \overline{OD} + t\vec{V}$. The

approach is recalled and the direction vector $\vec{V}$ is recalculated whenever the ray crosses an element face which connects two elements of different refractive indices. In fact, the deviation of rays at micro-scale is the laser diffusion at a macro-scale [34].

In previous studies [4,35,36], ray tracing method was used to perform simulations of the ray propagation in the semi-transparent part. This method allows to calculate the energy that reaches the weld interface and its location. Even though it is possible to couple the ray tracing method with Beer Lambert absorption law to determine laser energy lost during the propagation, the absorption in semi-transparent part is considered to be negligible, in this work.

4. Numerical procedure

Once a numerical composite geometry is constructed, it is discretised with an automated mesh generator on COMSOL Multiphysics. Then, numerical simulations are carried out to compute the deviation of the laser rays when transmitting through the semi-transparent medium.

The aim is to compare the diffusion patterns of laser rays, for different fibre orientation states. Three numerical geometries with different fibre orientation states are constructed (Fig. 8).

Out of the three numerical geometries, two (i.e. cases 2 and 3 in Fig. 8) are constructed using fixed fibre orientation tensors aiming to produce unidirectional fibre orientation with $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, and randomly

oriented fibres with $\begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$.

Given the incomplete information of fibre orientation tensor at a lower order, a new numerical geometry (case 1) is generated with

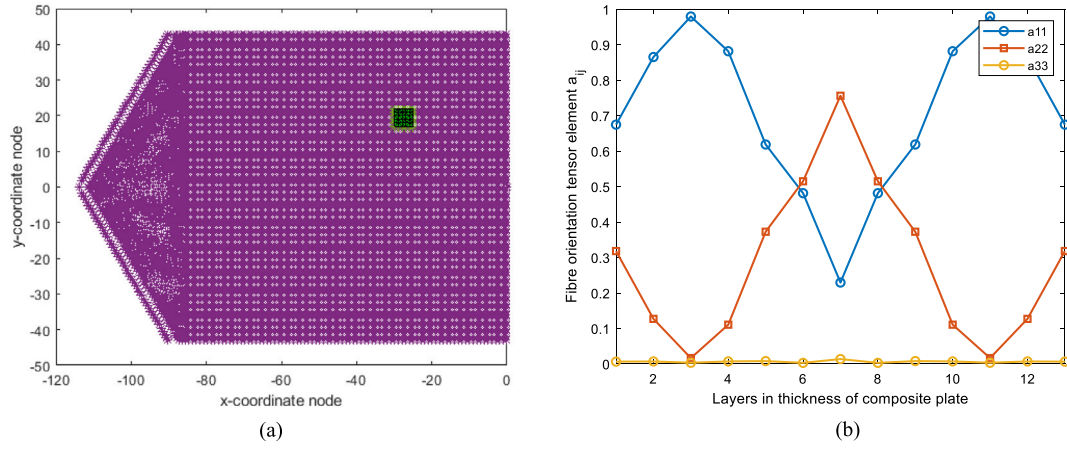


Fig. 11. (a) Location on injection moulded short-glass fibre reinforced PLA composite for fibre orientation distribution information, (b) Variation of homogenised fibre orientation distribution in thickness of sample plate.

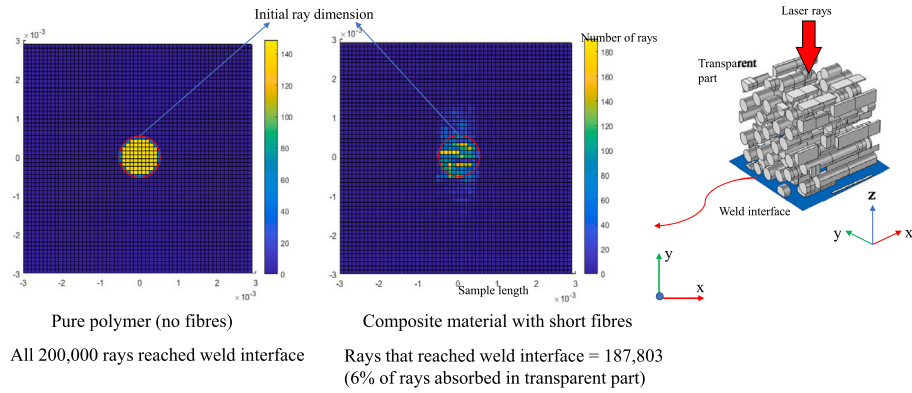


Fig. 12. Laser intensity distribution at weld interface.

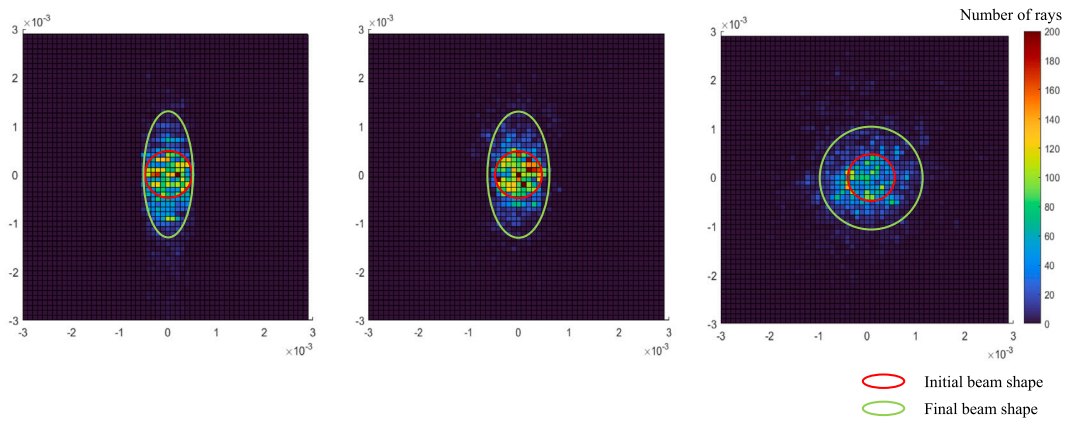


Fig. 13. Comparison of the three results.

perfectly unidirectional fibres. This implies that the tensor $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, that is expected to align the fibres parallel to axis 1, approaches the representation of case 1 only with very high order tensors. Referring to Fig. 4, these fibre inclusions are perfectly parallel to axis 1 with orientation angles $\theta = \pi/2$ and $\phi = 0$. Fig. 9 visualises the orientation of the fibre inclusions in each case of the numerical geometries.

The mean intensity of rays that reach the weld interface is obtained by gradually increasing the projected number of rays. A convergence is reached at 200000 rays (See Fig. 10).

Table 1

Numerical RVE specifications.

Fibre shape	Cylinder
Number of fibres (avg. in 3 cases)	79
Obstruction (avg. in 3 cases)	99%
Fibre volume fraction (avg. in 3 cases)	28%
Fibre aspect ratio (avg. in 3 cases)	7.1
RVE cube length	1500 μm
Grid size	50 μm

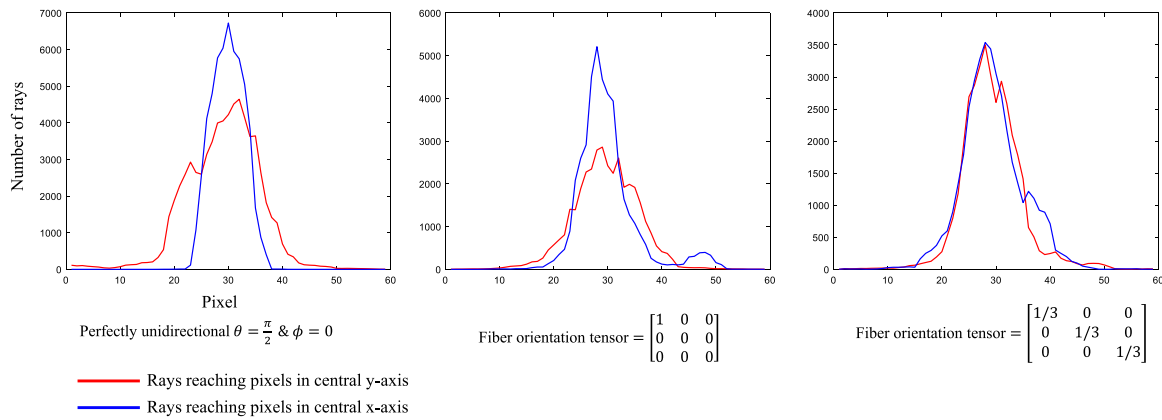


Fig. 14. Rays at the central lines of interface in x and y directions.

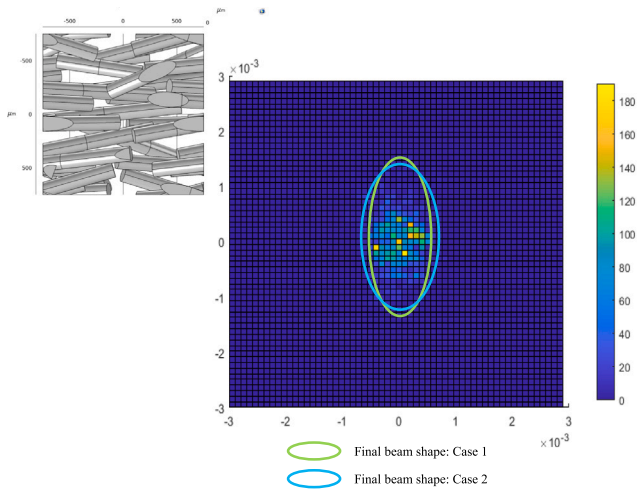


Fig. 15. Influence of fibre length on laser beam scatter.

Table 2

Ray tracing simulation specifications.

Laser beam diameter	1 mm
Number of rays	200000
Ratio of refractive index (n_{matrix}/n_{fibre})	1.44/1.55
Laser intensity distribution	Uniform distribution
Semi-transparent medium thickness	3 mm
Weld interface dimension	6 mm $\times$ 6 mm

The implemented ray tracing algorithm reduces the computation cost by using periodic geometries. The numerical geometries are repeated multiple times on all directions summing the thickness of the numerical geometry up to 3 mm and the interface surface to 6 mm $\times$ 6 mm plane. The non-focused laser beam diameter is 1 mm and the rays intensity is uniformly distributed. The ratio of refractive index of fibre and matrix is varied with 5% as proposed by Cosson et al. [34] as $n_m/n_f = 1.44/1.55$ (see Tables 1 and 2).

5. Experimental procedure

To experimentally validate the numerical modelling approaches, glass fibre reinforced composite plates are used. Two plates with 3 mm thickness similar to the numerically generated ones are chosen to project laser beam through the materials and the shapes of the laser beam diffused on the other surface is captured.

A uni-directional fibre reinforced composite plate is manufactured with glass fibres and polycarbonate (PC) by filament winding. Owing to

the manufacturing process, the fibre orientation is uni-directional [4] and suitably represents the numerical geometry of Case 1 in Fig. 8. The work also addresses the widening of laser beam perpendicular to fibre direction after being projected through the composite plate.

For the other cases, a short glass fibre reinforced Poly(lactic acid) (PLA) composite with thickness of 3 mm is manufactured using injection moulding. A simulation with the exact manufacturing conditions is performed on Autodesk Moldflow to determine the fibre orientation distribution. The information of fibre orientation tensors is extracted layer-wise along the thickness.

A location of 5 mm $\times$ 5 mm square on the simulated plate is found to exhibit an isotropic orientation distribution along the thickness of 3 mm (Fig. 11(a)). On this square area, each component of the fibre orientation tensor is averaged at each layer, i.e. at each thickness position. Fig. 11(b) represents average fibre orientation tensor components a_{11} , a_{22} and a_{33} values along the thickness of the plate in 13 layers.

The optical properties of the thermoplastics in the two plates can be easily obtained because both PC and PLA are completely transparent and are commonly used as transparent part during laser transmission welding [37,38]. The difference of refractive index of the fibre and matrix is close to 5% similar to the optical properties input in the numerical simulations. A laser module of 5 mW power with circular beam of 1 mm diameter is considered for experimental observations.

6. Results and discussion

The result of the ray tracing simulation is the number intensity of the rays and their location after the rays have travelled through the numerical geometry and reach the weld interface. This was demonstrated by projecting 200000 rays of a laser beam of 1 mm diameter into two numerical geometries of 3 mm thickness, one with no fibre inclusions (indicating a pure material) and the other with the fibre orientation tensor $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. A map of the diffusion of laser rays which have propagated through the two geometries is generated (Fig. 12).

When there are no fibres in the numerical geometry, the laser rays travel along straight paths showing no signs of scatter. On the contrary, when the rays travel in the composite, the ray paths are disturbed causing a deviation from their original paths. More importantly, it has been observed that the diffusion occurs in the direction perpendicular to the fibre inclusion orientation. Additionally, not all rays reach the target end, as they may have reflected back or refracted out from other faces of the numerical geometry. It is also possible that some of the rays circulated within the numerical geometry in an infinite loop.

Fig. 13 compares the diffusion map of 200000 rays for different orientation states. On an average, 160000 rays successfully reach the weld interface in all 3 cases. The other rays have either exited the

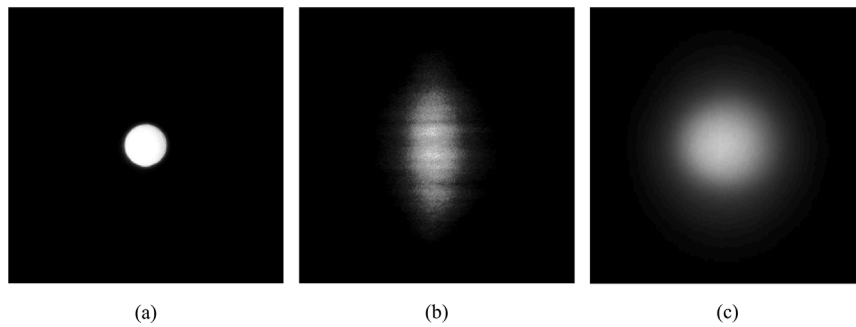


Fig. 16. Laser beam shape after propagating through (a) thin paper, (b) unidirectional glass fibre reinforced composite, and (c) randomly oriented glass fibre reinforced composite.

numerical geometry through the other faces or have refracted inside the numerical geometry in an infinite loop (which implies that the energy of these rays is completely absorbed in the numerical geometry; this case is not considered in this work). Additionally, the influence of 99% obstruction of rays is seen in some parts of Fig. 13 in red areas. There is a high concentration of laser rays in these areas which implies that 100% obstructed region is not achieved. The 100% obstruction region results in high fibre volume fraction and corresponds to high computation costs.

Analysing the distribution of rays that reach the exit face of the geometry, it is quite clear in Fig. 14, that the laser rays have scattered significantly. Especially in Case 1 (perfectly unidirectional orientation, Fig. 14(a)), the rays diffuse perpendicular to the fibre orientation. At the location of the fibre interaction, the rays have only two paths for refraction: either the perpendicular or the longitudinal to the fibre direction.

It is evident that the results for Case 1 and Case 2 should be very similar because the fibre orientation is similar for these two cases. It should be noted that the cylinder shape of the fibres has made the results explicit.

The influence of the low order tensors is quite evident when comparing the diffusion for Cases 1 and 2. Even if the simulation results are expected to be similar for these two cases, there is a notable difference due to the incomplete fibre orientation information contained in the fourth order tensor. The slight misalignment of the fibre inclusions have caused the diffusion in the direction of the fibre orientation. It is also possible that the laser rays reaching the ends of the inclusions are free to diffuse in various directions. One such encounter in a ray path could disturb the expected pattern of diffusion at the interface. To verify this, a numerical structure with longer fibre inclusions is constructed and the diffusion pattern is compared (Fig. 15). The laser ray diffusion is more compacted in the centre with less scatter in the direction of the fibre inclusions. Hence, the length of the fibres influences the diffusion of rays.

More disruptions of ray path lead to greater diffusion of rays. It may be concluded that the ray has propagated in the semi-transparent part by reflecting and refracting, resulting in a significant attenuation of energy before reaching the weld interface. This attenuation can be strongly dependent on the fibre volume fraction in the semi-transparent part. A high number of rays reaching the one location of the interface results in large absorption of energy locally which can lead to the degradation of the polymer. With a controlled initial beam intensity, it is possible to associate the fibre morphology such as fibre volume fraction, fibre shape, fibre length and fibre orientation to the optimal weld parameters such as laser beam intensity, laser diameter and laser beam speed. An increase in fibre volume fraction decreases the possibility for a laser beam to propagate through the material without scattering [39].

The simulations for Cases 1 and 3 were validated experimentally by projecting a laser beam through composites with similar fibre orientation. Fig. 16 shows these two cases with glass fibres.

The initial shape of the beam was observed by projecting the laser through a thin paper and a picture was captured from the other end with a digital camera (Fig. 16(a)). Replacing the paper with unidirectional and randomly oriented glass fibre reinforced composite plates, the laser beam shapes as shown in Fig. 16(b) and (c) are observed respectively.

Evidently, unidirectional fibre composite diffuses the laser beam in the direction perpendicular to the fibre direction, which corresponds to the ray tracing simulations result. It is also seen that random fibre orientation of fibres diffuses the laser in all directions which is also similar to the simulation result.

It should also be noted that the ratio of refractive index greatly influences the quantity of diffusion. The dimensions of the diffused laser spot can be altered by changing the fibre optical properties. Replacing the glass fibres by more opaque fibres such as flax, the extinction will be greater. In this case, the absorption of light has to be simulated in addition to the refraction [34]. The energy reaching the weld interface can be expected to be lower, even if the energy absorption in semi-transparent part is neglected.

7. Conclusion

The influence of the fibre orientation on the laser beam scattering was numerically modelled. The distribution of the laser energy that reaches the absorbent part is affected by the multiple refraction at each fibre/matrix interface. This effect was simulated by using the ray tracing method. This numerical method only needs the refraction indices ratio and the geometric representation of the microstructure. The initial distribution is spread and the overall intensity decrease with the thickness and the fibre volume fraction. A numerical model to generate microstructures representing composite materials considering fibre volume fraction, fibre shape, fibre length and fibre orientation was developed. Considering the influence of fibre orientation on both the mechanical properties and the laser welding quality, it is of great significance to take into account the real fibre orientation in the simulation of laser welding process. The simulated distribution of the laser can be used for FEM simulations to compute the temperature field at the weld interface, which is helpful for the optimisation of laser transmission welding of composites.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank European Regional Development Fund and the regions of Wallonia and Flandres for the financial support towards the project INTERREG FWVL ATHENS.

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